6 multiple access protocols

Problem 6.1 : a slotted ALOHA has $N = 10$ nodes. each node has a backlogged packet and independently attempts to transmit in each slot with probability p

a) write an expression for the probability that exactly one node transmits in a given slot

hint : slotted aloha : $P(\text{success}) = Np(1-p)^{N-1}$ → $P(\text{success}) = 10p(1-p)^9$

success = 1 sent in

b) use the continuous approximation $S = Ge^{-G}$ where $G = Np$, find the value of p that maximizes throughput for $N = 10$

hint: maximized @ $G = 1$

$G = Np$

$1 = 10p$

$p = \frac{1}{10}$

c) what is the maximum throughput S_{max} in slots/slot ? in percentage

$S_{max} = Ge^{-G}$ → max @ $G = 1$ → $S_{max} = 1e^{-1} = \frac{1}{e} = 36.8\%$

d) how does this compare to pure ALOHA maximum throughput

pure aloha throughput $= S = Ge^{-2G}$ @ $G = 0.5$ → 18.4%

↳ slotted aloha is much more efficient → twice

Problem 6.3 : two stations A and B both attempt to transmit and collide for the first time (collision count $K = 1$)

a) from what range does each station independently choose a back off value?

after $K = 1$ → $[0, 2^1 - 1]$ → back off → time waited after colliding in CD, $(0, 2^{\min(K, 10)} - 1)$

capped out at 1024 slots so you dont wait forever

$[0, 1]$

do they choose wait time randomly?

b) what is the probability they collide again at the next step?

↳ collide if both choose same backoff time:

$P(\text{collide}) = P(\text{both } 1) + P(\text{both } 0) = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2}$

each picks a random time from $(0, 1, 2 \ldots$ to wait before retransmitting

c) after a second collision ($K = 2$) what is the back off range

$\{0 \ldots 2^{\min(2, 10)} - 1\}$ → $\{0, \ldots, 3\}$

$[0, 1, 2, 3]$

d) after how many successive collisions does CSMA/CD give up ? what happens then ?

$K = 16$ → send message to higher up